MIL-A-36188A
6 August 1974
SUPERSEDING
MIL-A-36188 (DM)
10 April 1964

MILITARY SPECIFICATION

APRON, DENTAL, PLASTIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a plastic apron suitable for use as a protector from debris, tissue exudates and water used in dental operative procedures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

L-P-375	Plastic Film, Flexible, Vinyl Chloride.
V-T-276	Thread, Cotton.
DDD-T-86	Tape, Textile, Cotton, General Purpose (Unbleached, Bleached, or Dyed).
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-F-21840	Fastener Tapes, Hook and Pile, Synthetic.

STANDARDS

Federal

FED-STD-406	Plastic, Methods of Testing.
FED-STD-751	Stitch, Seam, and Stitching.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 6532

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DRAWING

Defense Personnel Support Center (DPSC)

24347 APRON, DENTAL, PLASTIC

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 OTHER PUBLICATION. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES DEPARTMENT OF COMMERCE
- NATIONAL BUREAU OF STANDARDS

Commercial Standard CS 192-53
General Purpose Vinyl Plastic Film

(Application for copies should be addressed to the United States Department of Commerce, National Bureau of Standards, Washington, D. C. 20234).

3. REQUIREMENTS

3.1 Material.

3.1.1 Vinyl plastic. The plastic sheeting shall be a flexible vinyl waterproof film conforming to type I of L-P-375, milky white or opaque in color, with a smooth finish, except that the following characteristics shall not apply: ultimate elongation, resistance to weathering, cold crack, and lacquer lifting. Blocking shall be determined as specified in method 1131 of FED-STD-406 at 70 ± 1 degrees centigrade (°C.). The plastic sheeting shall meet the flammability requirements specified in 3.1.1.3.

3.1.1.1 Apron body. The apron body sheeting shall be 4 gage (0.004 inch) plus or minus 10 percent plastic sheeting (see 3.1.1).

3.1.1.2 Neckband. The neckband sheeting which forms the collar shall be 12 gage (0.012 inch) plus or minus 10 percent plastic sheeting (see 3.1.1).

3.1.1.3 Flammability. The rate of burning shall not exceed 1.2 inches per second as judged by the average of five determinations transverse to the directions of processing, when the material is tested as specified in 4.4.1.

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3.1.2 Nylon tape fastener. The nylon tape fastener shall be white, type I, class 1 hook and pile in accordance with MIL-F-21840.

3.1.3 Cotton.

3.1.3.1 Cotton binding. The cotton binding used for finishing the edges of apron, except for neck closure, shall be preshrunk, white cotton tape, type I, or type III, class 2, of DDD-T-86. As an alternate, for binding the edges of apron, bleached cotton sheeting of 100/60 designation, conforming to a good commercial quality, may be used.

3.1.3.2 Thread. The thread shall be white, 3-ply cotton, type IA2, ticket No. 40 or 50, or type IC1, ticket No. A or 0, in accordance with V-T-276.

3.2 Design and construction. The apron shall be suitable for patient or operator use during dental operative procedures. The plastic apron shall consist essentially of an apron body having all edges, except neck hole opening, reinforced by machine sewn cotton binding; two tie tapes at the waistline; a centrally located keyhole type neck opening; reinforced with 2-ply vinyl plastic sheeting (see 3.2.1 and 3.2.1.2) forming a collar, with an adjustable neck closure to accommodate a range of neck sizes.

3.2.1 Apron body. The apron body shall be made of one-piece flexible vinyl plastic film as specified in 3.1.1 and 3.1.1.1. The apron body shall conform to the design and dimensions shown in Defense Personnel Support Center (DPSC) Drawing 24347.

3.2.2 Cotton binding. All edges of the apron body, with the exception of the neck line, shall be reinforced by white, bleached cotton tape or sheeting as specified in 3.1.3.1. The cotton binding shall terminate at the ends of neckband and shall be folded over on itself and be bartacked by a minimum of 42 stitches. The cotton binding, made of one (1) inch wide stock tape or sheeting, shall be machine sewn with white cotton thread (see 3.1.3.2), using a 301 or 401 class stitch in a BSc-1 type seam, in accordance with FED-STD-751. The seam shall have 8 to 12 stitches per inch. The dimensions of finished binding shall be as shown in DPSC Drawing 24347.

3.2.3 Ties. A tie shall be securely fastened to each side of the apron body by means of a 42 stitch bartack, using cotton binding specified in 3.1.3.1. The cotton binding shall be machine sewn as described in 3.2.2, except that an SSe2b type seam shall be used. The free ends of the tie tapes shall be finished to prevent ravelling. The dimensions of the ties shall be as shown in DPSC Drawing 24347.

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3.2.4 Neckband. The neck opening, centered in the top end of the apron body shall be finished with a 2-ply plastic material as specified in 3.1.1.2, forming a stand-up, open, circular collar. The 2-plies shall be machine sewn together using white cotton thread conforming to 3.1.3.2, with a 401 class stitch in an SSe-2b type seam for the upper neckband edge and an SSq-2b type seam for the lower neckband edge, in which the apron body neckline material shall be included as the middle ply. Stitches and seams shall be in accordance with FED-STD-751. The finished, attached neckband shall conform to the dimensions shown in DPSC Drawing 24347. The neckband shall have a hook and pile adjustable nylon fastener tape closure as specified in 3.1.2. The hook shall be machine sewn onto the left outside face of the neckband. The pile shall be machine sewn onto the right inside face of the band. The hook and pile shall be machine sewn using cotton thread conforming to 3.1.3.2, with 14 to 16 class 301 or 401 stitches per inch in accordance with FED-STD-751. The dimensions and positioning of the hook and pile shall be as shown in DPSC Drawing 24347.

3.3 Laundering. The finished apron shall be suitable for laundering with lukewarm water and mild soap when tested as specified in 4.4.2. The apron shall remain flexible, soft, colorless, and intact, showing no signs of stiffening, shredding, tearing or disintegrating. The marking on the identification label shall remain legible and permanent after laundering.

3.4 Identification marking. A label shall be sewn into the bottom left corner of the binding of the apron. The label shall be permanently marked with the manufacturer's name or registered trademark.

3.5 Workmanship. Workmanship shall be first class throughout. The finished aprons shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of compliance. Certificates of compliance, supplied by the manufacturer of the raw materials, may be furnished in lieu of actual performance of such testing by the contractor. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I, III and IV (see 4.2.1, 4.3.1, 4.3.2, and 4.3.3) and table II (see 4.2.2).

4.2.1 Sampling for examination of apron defects. Sampling for visual and dimensional examination (see 4.2) shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) expressed in defects per hundred units shall be as specified in table I. Unit of product for examination shall be one apron when the examination is performed on the end item; unit of product for examination shall be the applicable component part of the apron when the examination is performed on a component part basis.

Table I. Sampling for Examination

	Inspection levels	AQL's Major	Total
For defects applicable to 4.3.1	I	4.0	10.0
For defects applicable to 4.3.2	S-1	One class	6.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. The inspection levels and AQL's shall be as indicated in table II. Unit of product for tests performed on end item shall be one apron (unit of issue). When tests are performed on raw material, a sufficient amount of each lot of raw material used shall be taken as the sample unit of product.

Table II. Tests II.

Characteristic	Requirement	Test Method	No. of determinations per sample unit	Results reported as	Insp. level	AQL
Flammability 1/	3.1.1.3	4.4.1	2	Pass or fail	S-1	6.5
Laundering	3.3	4.4.2	1	Pass or fail	S-1	2/

1/ Flammability tests may be performed either on the raw material or on the end item.

2/ Acceptance number shall be zero for all sample sizes.

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4.3 Examination of apron defects. The defects found during examination shall be classified in accordance with 4.3.1 and 4.3.2.

4.3.1 Examination of apron, dental, plastic for visual characteristics.

NOTE: Defects shown as an asterisk (*) are to be classified as indicated below:

Major defect - when seriously affecting serviceability or appearance
Minor defect - when affecting serviceability or appearance, but not seriously

Table III. Classification of Apron Defects

EXAMINE	DEFECT	MAJOR	MINOR
Quality and finish of plastic material	Material other than specified	X	
	Material not color specified	(*)	
	Bubble, sharp crease, wrinkle, blister, gouge or other imperfection	(*)	
	Cut, tear, hole, break or crack	X	
	Finish not as specified		X
	Objectionable odor (other than odor typical to Vinyl Chloride)		X
Design	Any characteristic not in accordance with the specified requirements or standard sample	X	
Construction and Workmanship			
General	Apron body not one-piece construction	X	
	Component missing	X	
	Required operation omitted or not properly performed	X	
	Foreign matter or material imbedded in the plastic sheeting	(*)	
	Seam misplaced, or out of alignment, by more than 1/4 inch	(*)	
	Any component or part misaligned, malformed, or damaged	(*)	

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Table III. Classification of Apron Defects (Cont'd)

EXAMINE	DEFECT	MAJOR	MINOR
Seams and stitching	Seam type not as specified	X	
	Stitch type not as specified	(*)	
	More than maximum number of stitches per inch damaging material	X	
	More than maximum number of stitches per inch but not damaging material		X
	One stitch per inch less than specified minimum		X
	Two or more stitches per inch less than the specified minimum	X	
	Any stitching omitted where required	(*)	
	Loose stitch and loosely secured seam	(*)	
	Open seam-one or more stitches broken or 2 or more continuous stitches skipped on single stitch seam or on both rows of double stitch seam	X	
	Seam puckered or distorted	(*)	
	One bartack missing, loose, or misplaced	(*)	
	Two bartacks missing, loose, or misplaced	X	

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Table III. Classification of Apron Defects (Cont'd)

EXAMINE	DEFECT	MAJOR	MINOR
Ties	Less than required finished length	X	
	Free end not properly finished	X	
	Soiled permanently		X
	Not secured adequately	X	
	Width not as specified	(*)	
Neckband	Not double thickness of plastic	X	
	Twisted in attachment to apron body so that it does not lie flat on neck	X	
	Ends of plastic band not completely bound by reinforcing cotton tape	X	
Fastener tape	Hook and pile not of type specified	X	
	Hook sewn on inside surface of neckband	X	
	Hook and pile do not function as specified	X	
	Hook and pile not positioned as required	X	
	Hook and pile not mounted as required	X	
	Hook or pile becomes detached during operation	X	
Identification marking	Omitted, incomplete, illegible, incorrect, or misplaced (marking defects scored once per apron regardless of number of faults)		X

4.3.2 Examination for dimensions. Any dimension that is not within the specified tolerance shall be classified as a defect. Examination for dimensional compliance may be conducted on the end item or component parts.

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4.3.3 Examination of preparation for delivery requirements. An examination shall be made to determine that applicable packaging, packing and marking requirements are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closures listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

Table IV. Classification of Packaging Defects

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application
Material	Any component missing
	Any component damaged, affecting serviceability
Workmanship	Inadequate application of components, such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling
	Bulging or distortion of containers
Weight or content (exterior and interior)	Number of intermediate packages or bundles is more or less than required; gross/net weight exceeds requirements

4.4 Tests.

4.4.1 Flammability. The flammability of the material used in the apron shall be tested as specified in Federal Standard Method 4.10 of U. S. Department of Commerce Standard 192-53 for general purpose vinyl plastic film.

4.4.2 Laundering test. The apron shall be immersed in a minimum of 2 gallons of one (1) percent neutral toilet soap solution maintained at 122 $\pm$ 2 degrees fahrenheit (°F.) (50° $\pm$ 1.1° C.) for a period of one hour. The apron shall then be alternately dipped into and raised out of the soap solution for a period of 5 minutes, repeating the dipping cycle a minimum of 10 times per minute. This process shall be performed so as to assure exposure of entire apron surface to the soap solution. Upon removal after the test period, the sample shall be rinsed with warm water and shall be examined for characteristics specified in 3.3.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. Each apron, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	72 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each apron shall be neatly folded and inserted into a clear, snug fitting polyethylene bag having a minimum thickness of not less than 0.00125 inch. A 1/4 inch diameter hole shall be made at one corner of the bag to allow excess air to escape and the closure shall be effected by heat sealing or by means of a tuck or reverse flap.

5.1.3.2 Intermediate package. Intermediate package shall be a snug fitting box constructed in accordance with PPP-B-676 except that commercial colors are acceptable or PPP-B-636, type CF, class domestic. Closure shall be secured to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. The exterior container shall also conform to the special requirements as specified in PPP-B-636. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

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NOTE: Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage or redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. The exterior container shall also conform to the special requirements as specified in PPP-B-636. Closure shall conform to method I or II as specified in the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number, size, and name of contractor shall be shown.

NOTE: Marking may be printed directly on the bag or on a label inserted within the bag in such a manner so as to permit ready identification.

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 Specification changes. Because of the extensiveness of changes to this document, the margins have not been marked with an asterisk to indicate where changes from the previous issue have been made.

6.3 National Stock Number (NSN) Coverage. This specification covers the following item appearing in the Federal Supply Catalog:

Stock No.	Item Identification
6532-00-890-1429	APRON, DENTAL, Plastic: 45 in. long, 31 in. wide. Adjustable neckband closing at back.

6.4 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6532-1083

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-A-36188A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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